Section 3.8 Exercises

Directions: For exercises 1 - 6, find $f^{-1}(x)$ for each function.

1. $f(x) = x + 3$

2. $f(x) = x + 5$

3. $f(x) = 2 - x$

4. $f(x) = 3 - x$

5. $f(x) = 11x + 7$

6. $f(x) = \frac{2x+3}{5x+4}$

Directions: For exercises 7 - 9, find a domain on which each function f is one-to-one and non-decreasing. Write the domain in interval notation. Then find the inverse of f restricted to that domain.

7. $f(x) = (x + 7)^2$

8. $f(x) = (x - 6)^2$

9. $f(x) = x^2 - 5$

10. Given $f(x) = \frac{x}{2+x}$ and $g(x) = \frac{2x}{1-x}$:

ⓐ Find $f(g(x))$ and $g(f(x))$.
ⓑ What does the answer tell us about the relationship between $f(x)$ and $g(x)$?

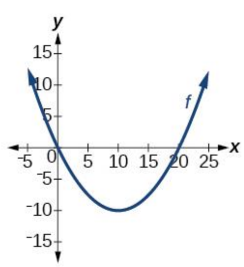
Directions: For exercises 11 - 12, algebraically verify that $f(x)$ and $g(x)$ are inverses.

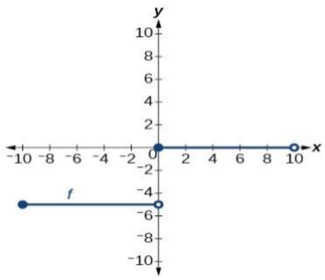
11. $f(x) = \sqrt[3]{x - 1}$ and $g(x) = x^3 + 1$

12. $f(x) = -3x + 5$ and $g(x) = \frac{x-5}{-3}$

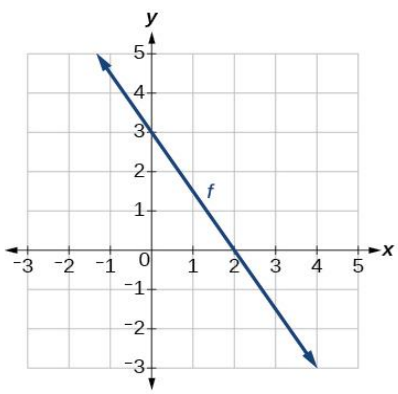
Directions: For exercises 13 - 14, determine whether the graph represents a one-to-one function.

13.



14. 

Directions: For exercises 15 - 18, use the graph of f shown below.



15. Find $f(0)$.

16. Solve $f(x) = 0$.

17. Find $f^{-1}(0)$.

18. Solve $f^{-1}(x) = 0$.

Directions: For exercises 19 - 26, find the inverse of the function on the given domain.

19. $f(x) = (x-4)^2,\ [4, \infty)$

20. $f(x) = (x+2)^2,\ [-2, \infty)$

21. $f(x) = (x+1)^2 - 3,\ [-1, \infty)$

22. $f(x) = 2 - \sqrt{3+x}$

23. $f(x) = 3x^2 + 5,\ (-\infty, 0]$

24. $f(x) = 12 - x^2,\ [0, \infty)$

25. $f(x) = 9 - x^2,\ [0, \infty)$

26. $f(x) = 2x^2 + 4,\ [0, \infty)$

Directions: For exercises 27 - 42, find the inverse of the functions.

27. $f(x) = x^3 + 5$

28. $f(x) = 3x^3 + 1$

29. $f(x) = 4 - x^3$

30. $f(x) = 4 - 2x^3$

31. $f(x) = \sqrt{2x+1}$

32. $f(x) = \sqrt{3-4x}$

33. $f(x) = 9 + \sqrt{4x - 4}$

34. $f(x) = \sqrt{6x - 8} + 5$

35. $f(x) = 9 + 2\sqrt[3]{x}$

36. $f(x) = 3 - \sqrt[3]{x}$

37. $f(x) = \frac{2}{x+8}$

38. $f(x) = \frac{3}{x-4}$

39. $f(x) = \frac{x+3}{x+7}$

40. $f(x) = \frac{x-2}{x+7}$

41. $f(x) = \frac{3x+4}{5-4x}$

42. $f(x) = \frac{5x+1}{2-5x}$